

HEAR: Homomorphically Encrypted Allreduce

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ABSTRACT

Allreduce is one of the most commonly used collective operations. Its latency and bandwidth can be improved by offloading the calculations to the network. However, no way exists to conduct such offloading securely; in state-of-the-art solutions, the data is passed unprotected into the network. Security is a significant concern for High-Performance Computing applications, but achieving it while maintaining performance remains challenging. We present HEAR, the first high-performance system for securing in-network compute and Allreduce operations based on homomorphic encryption. HEAR implements carefully designed and modified encryption schemes for the most common Allreduce functions and leverages communication domain knowledge in MPI programs to obtain decryption and encryption routines with high performance. HEAR operates on integers and floats with no code base and no or little hardware changes. We design and evaluate HEAR, showing its minimal overhead, and open-source our implementation. HEAR represents the first step towards achieving confidential HPC.

CCS CONCEPTS

• Networks → In-network processing; • Security and privacy → Distributed systems security; • Computer systems organization → Distributed architectures.

KEYWORDS

Message Passing Interface, Allreduce, In-Network Computing, Homomorphic Encryption, Confidential Computing

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1 INTRODUCTION

Allreduce is the most used [21, 62, 76, 10] message-passing interface (MPI) collective in which P processes belonging to a communicator $\mathbb{S}$ aggregate a vector of data element-wise and return the aggregated values to each process. Summation, product, and max/min operators

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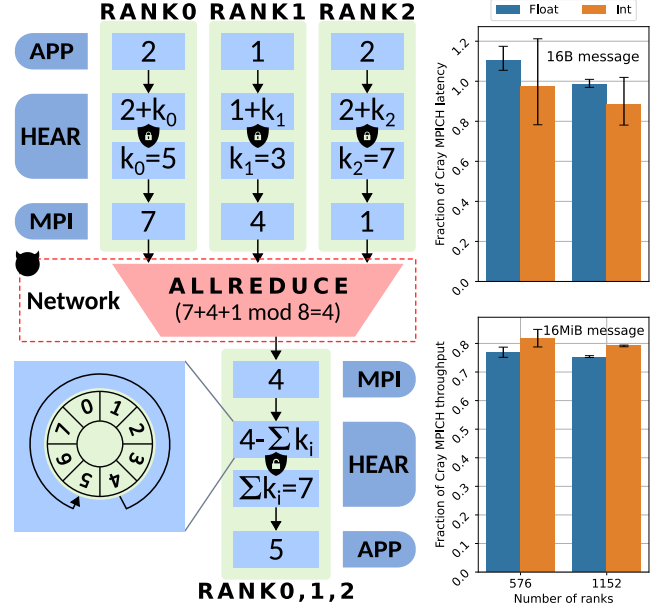


Figure 1: A simplified overview of HEAR running within a secure environment with an untrusted network. HEAR conducts homomorphically encrypted Allreduce by shifting the values on a ring, ensuring losslessness and enabling performant decryption through spatial knowledge of the program.

are typical examples of aggregation functions [64]. Many applications, such as scientific workloads [21, 44, 89], deep learning [88, 20, 8], graph processing [100, 51], and big data analytics [100, 101], rely on Allreduce. Recent studies show that Allreduce, together with the related Reduce collective, are the most commonly invoked collectives taking over 30-40% of the total collective operation time in core hours [21, 76] and making “MPI_Allreduce [...] the most significant collective in terms of usage and time”.

To increase the performance and lower the amount of transmitted data of Allreduce, the processes can exploit in-network computing (INC). INC allows hosts to offload the Allreduce function to the networking devices such as switches [45, 60, 84] or network interface cards (NICs) [86, 50]. INC provides two critical advantages: latency and bandwidth usage reductions. The former is lowered by 3-18x [45, 60] which results in performance gains of 1.5-5.5x [60, 84]. The latter can be reduced by 2x [27], resulting in a lower network contention and system power savings as modern datacenter interconnects constitute 15-50% of the overall power usage [2].

However, INC is insecure, and currently, no method exists for securing it. As the information required for computations is used unencrypted, the state-of-the-art sends the INC data unprotected